

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Ship 10+ production TypeScript features: document pipelines (PDF gen, Microsoft Graph API email parsing, S3), multi-model AI integrations (Claude, GPT-4o, Gemini), Auth0 RBAC; 923 commits, 7 deployments. - Own full-stack TypeScript/Node.js services: PostgreSQL and Redis data layer, Docker containerization, Auth0 RBAC, Sentry error tracking, and Temporal.io agentic workflow orchestration for reliability.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling interactive visualization of 500GB+ strawberry and grape genome datasets for 100+ researchers. - Improved ML model accuracy by 12% by applying hyperparameter tuning and time-series feature engineering on biological sensor data collected from strawberry and grape genomic field experiments.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation using OCR, NLP, and ML classification pipelines on medical records; improved end-to-end document processing accuracy by 25% across 10K+ daily claims. - Automated cloud deployments across AWS and GCP via CI/CD; integrated Grafana observability dashboards and led model validation and red-teaming sessions to improve reliability across production services.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for a data-driven enterprise ticket analysis system using NLP classification models; reduced average resolution time by 15% and eliminated manual triage for 2,000+ monthly tickets. - Implemented distributed data pipelines using Kafka streaming and Spark + Hadoop batch processing for large-scale ticket ingestion; reduced end-to-end ticket-processing latency by 20% at peak load.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.		

Projects

VectorLift	Semantic Search & Ranking Engine	2026
- Built BM25 + bi-encoder (sentence-transformers) + cross-encoder re-ranker pipeline on MS MARCO with 6 configurations evaluated across NDCG@10, MRR@10, MAP, and bootstrap significance testing. - Indexed 8.8M-passage MS MARCO corpus via Kafka into Elasticsearch and Qdrant vector stores; built FastAPI search API and Streamlit experiment dashboard comparing dense vs. sparse retrieval performance.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard and activity feed in TypeScript with NestJS and Next.js; MySQL transactional writes, MongoDB feed projections with TTL, Redis sorted sets for sub-millisecond reads; live SSE. - Validated multi-DB consistency with Testcontainers integration tests; benchmarked hot paths with k6 at 10K concurrent users; wired OpenTelemetry + Prometheus observability and GitHub Actions CI/CD pipeline.		

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Bash
- Frontend & Full Stack:** React, Next.js, Node.js, NestJS, REST APIs, Auth0, PostgreSQL, Redis, MongoDB, SSE
- Search & AI:** Elasticsearch, Qdrant, sentence-transformers, Claude/GPT-4o/Gemini APIs, LangChain, RAG, Temporal.io
- DevOps & Cloud:** Docker, Kubernetes, AWS, GCP, GitHub Actions, Kafka, Prometheus, Grafana, Sentry

Additional Experience and Awards

- IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete full-stack MVPs end-to-end in 48-hour competitive sprints, each delivering a working live demo.